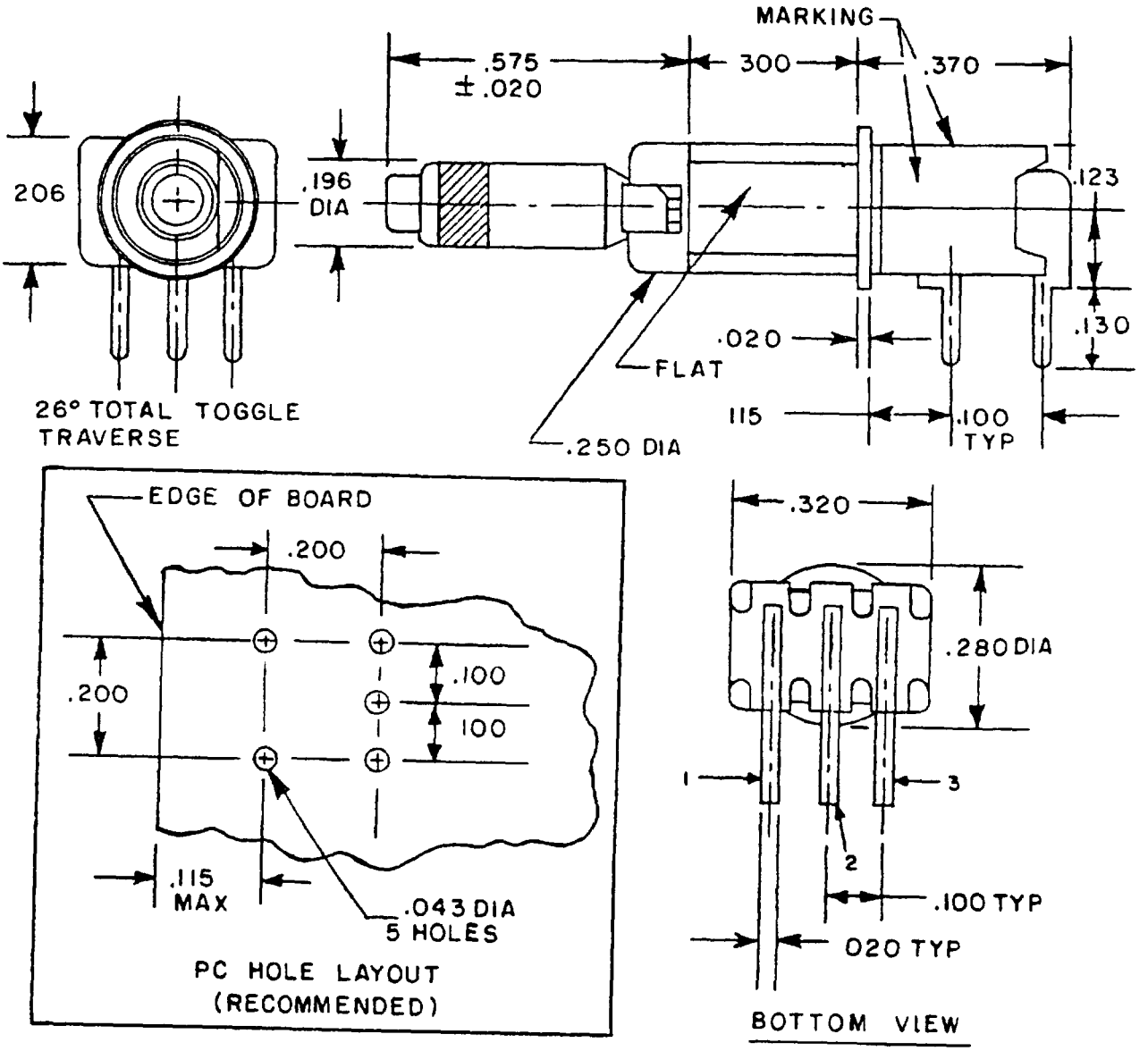
MILITARY SPECIFICATION SHEET

SWITCHES, TOGGLE, SUBMINIATURE, LEVER LOCK, SEALED LEVER, FLUX SEALED RIGHT ANGLE PC BOARD MOUNT - ONE, TWO, AND FOUR POLE, LOW LEVEL CONTACTS

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-83731



Configuration A
Single pole

FIGURE 1. Dimensions and configurations

(A) denotes changes

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:33Z
Check the source to verify that this is the current version before use.

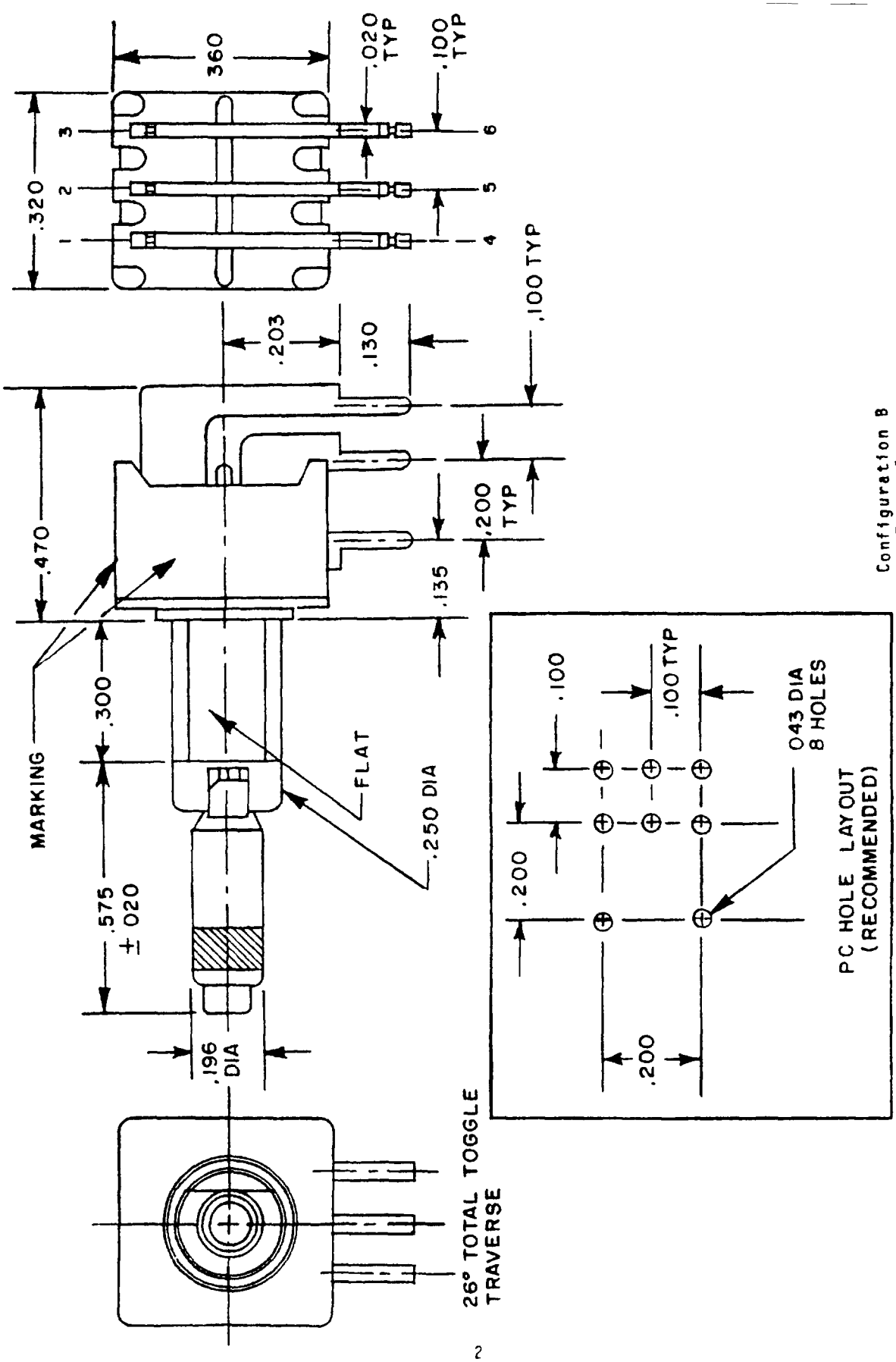


FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:33Z
Check the source to verify that this is the current version before use.

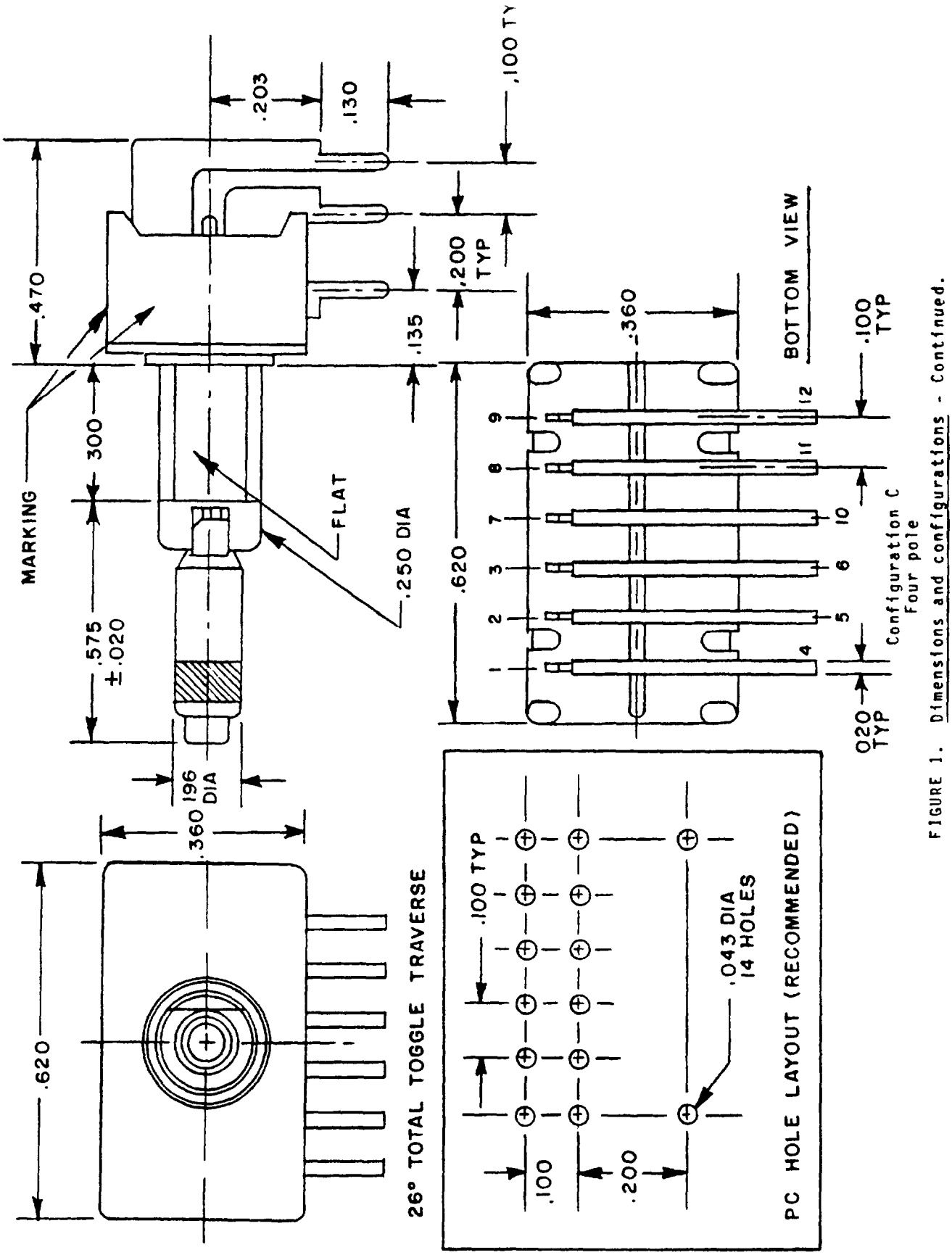


FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:33Z
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm
.020	0.51	.206	5.23
.043	1.09	.250	6.35
.100	2.54	.280	7.11
.115	2.92	.300	7.62
.123	3.12	.320	8.13
.130	3.30	.360	9.14
.135	3.43	.370	9.40
.196	4.98	.470	11.96
.200	5.08	575	14.61
.203	5.16	.620	15.75

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for information only
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) and ±4°.
4 Action of internal mechanism movement is opposite to direction of toggle movement.
5. Terminal numbers are for reference only and may not appear on switch

FIGURE 1. Dimensions and configurations - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:33Z
Check the source to verify that this is the current version before use.

TABLE I. Circuit for switching characteristics.

Circuit 1/	Available 1/ locking combinations	Circuit with toggle in		
		Flat side	Center	Opposite flat side
21	A	On	Off	On
23	D	On	None	On

1/ Circuit numbers and locking combinations are derived from MS27740(USAF).

TABLE II. Dash numbers and characteristics.

M83731/20 Dash no.	No. of poles	Circuit 1/ configuration	Applicable figure
-001	1	21	1A
-002	2	21	1B
-003	4	21	1C
-004	1	23	1A
-005	2	23	1B
-006	4	23	1C

1/ Circuit configurations are described by table I.

REQUIREMENTS:

Dimensions and configurations See figure 1 and configurations A through C as applicable

Material and finish:

Contacts.

Moving: 40 microinches minimum gold thickness over 80 microinches minimum nickel thickness over copper alloy.

Stationary Gold-plate (40 microinches minimum thickness) over nickel-plate (80 microinches minimum thickness) over copper-alloy.

Terminals Gold-flash (8 microinches minimum) over nickel-plate (80 microinches minimum) over copper-alloy.

Weight (maximum): 1 pole, 4.25 grams, 2 pole, 5 75 grams, 4 pole, 6.75 grams.

Temperature rating: -25°C to +71°C (operating), -55°C to +85°C (storage).

Electrical ratings:

Low Level to .4 VA (with voltage limited to 20 V dc)

Low Level test limits defined as 10 mA maximum at open circuit voltage of 30 mV maximum (ac or dc)

Mechanical endurance Except temperatures of cycling shall be -25°C, +0°C, -4°C and +71°C, +4°C, -0°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:33Z
Check the source to verify that this is the current version before use.

Electrical endurance (except as follows)

Low level: Per MIL-STD-202, method 311, using an open circuit test voltage of 30 mV (ac or dc) and a closed circuit current of 10 mA. 20,000 operating cycles at 20 cycles per minute minimum. No miss resistance in excess of 3 ohms Contact sticking not applicable.

Overload. Not applicable.

Short circuit: Not applicable.

Shock: Method I.

Terminal flux seal: These switches shall have their bases sealed to prevent flux from entering the switches during soldering processes. They shall be tested for this capability as follows: Measure and record initial contact resistances. Support switches, terminals down, in a shallow pan. Pour flux at 80°F ± 5°F per type RMA of MIL-F-14256, specific gravity 0.896 into pan without splashing until the level of flux is approximately 1/16 inch above bottom of plastic switch case and let switches soak for 10 minutes. Remove switches from flux, clean with flux cleaning solvent and immediately place in oven for drying at 175°F ± 10°F for two hours. After switches have cooled to room temperature, repeat measurement of contact resistance. Contact resistance shall not increase by more than 10 milliohms over the initial reading. Disassemble switch and visually inspect the contact area for evidence of flux. Any evidence of flux shall be cause for rejection.

(A) Sand and dust. Not applicable.

(A) Explosion test Not applicable.

Dielectric withstanding voltage:

Atmospheric pressure. 1000 V rms.

Reduced barometric pressure (10,000 feet). 250 V rms.

Strength of terminals:

Solderable terminals: Applicable.

Printed circuit (PC) terminals Shall be tested in accordance with method 211 of MIL-STD-202.

a. Test condition B.

b. For right angle switches, the bend shall be applied to the terminal portion that is inserted into a printed circuit board.

Part number The part number shall consist of the prefix M83731/20 followed by the appropriate three digit dash number from table II (e.g., M83731/20-001).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:33Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian
Air Force - 85

Review activities:
Air Force - 11, 99

Preparing activity
Air Force - 85

Agent:
DLA - ES

(Project 5930-F648-02)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T09:33Z
Check the source to verify that this is the current version before use.